Experiment 3

Average Sale Price of Houses with More Than Four Bedrooms

Aim

To calculate the average sale price of houses having more than four bedrooms using NumPy arrays and conditional filtering.

Procedure

1. Import the NumPy library.
2. Load the house dataset from a CSV file into a NumPy array named house_data.
3. Arrange the columns in the order of bedrooms, square footage, and sale price.
4. Use house_data[:, 0] > 4 to select houses with more than four bedrooms.
5. Store the filtered rows in a new array named selected.
6. Extract the sale prices using selected[:, 2].
7. Use np.mean() to calculate the average sale price.
8. Display the calculated average sale price.

Code

```
import numpy as np

house_data = np.array([
    [3, 1200, 250000],
    [5, 2000, 450000],
    [4, 1500, 320000],
    [6, 2500, 550000],
    [5, 2200, 500000]
])

selected = house_data[house_data[:, 0] > 4]

avg_price = np.mean(selected[:, 2])

print("Average sale price:", avg_price)
```

Result

Thus, the houses having more than four bedrooms were selected successfully, and their average sale price was calculated using NumPy array operations.

Output

```
========== RESTART: D:/fundamentails of data science/experiment 3.py ==========
Average sale price: 500000.0
```